PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

CHAMWINO DC

FORM FOUR MID TERM JOINT EXAMINATION

041 BASIC MATHEMATICS

TIME: 2:30 HOURS

INSTRUCTIONS

This paper consists of Section A and Section B.

Answer ALL questions in both sections.

Show clearly all working.

Use logarithmic tables and calculator where necessary.

SECTION A

(Answer ALL questions in this section)

1. (a) If $x * y = xy + x + y$, solve $x * x = 48$

(b) If $a * b = \sqrt{ab}$, find $2 * 18$.

2. (a) Approximate 0.003467 to 2 significant figures.

(b) Write 3,459,700 in standard form under two decimal places.

3. (a) Given $x = \sqrt{\frac{s-a}{s-b}}$, transpose and make **s** the subject.

(b) Given $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ make **u** the subject of the formula.

4. (a) Solve using logarithm tables: $\sqrt{0.6274}$

(b) Find x when $\log x = \bar{1}.4923$.

5.(a) Find the arithmetic mean of 10 and 40.

(b) Find the geometric mean of 2 and 18.

6. (a) It is given that sin A = 0.6 . Find A for $0 \leq A \leq 360$

(b) Given sin B = $\frac{m}{n}$, find cos B .

7. (a) Given f(x) = sin x , by use of graph find the domain and range of f(x) .

(b) Given f(x) = $\frac{3}{x^2+2}$ find $f^{-1}(1)$

8.(a) Solve the following system using matrices:

$$4x + 5y = 2$$
$$2x + 3y = 0$$

(b) If A =$\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$, find A^2

9. (a) If log x + 2 log y = 1 , find x in terms of y .

(b) Use logarithm tables to find x when $17^X = 34$.

10. (a) Factorize 1 - $9x^2$.

(b) Solve 10 - 7x - $12x^2$ = 0 .

SECTION B

Answer All Questions In This Section

11. (a) Two football matches were played. Find the probability of:

(i) All win

(ii) No win

(b) A family is planning to have two children. Find the probability of getting:

(i) All girls

(ii) A boy and a girl

12. The table below shows marks obtained by students:

Marks	0–10	10–20	20–30	30–40	40–50
Number	4	5	10	2	1

(a) Find:

(i) Mean

(ii) Mode

(iii) Median

(b) Sketch the histogram.

13. The eleventh term of an A.P. is 40 and the fifteenth term is 56.

(a) (i) Find the tenth term.

(ii) Find the sum of the first twentieth terms.

(iii) Find the sum of the first nth terms.

(b) Show that for $r < 1$, the sum of the first n terms of a G.P. is $\frac{G_1\,(1 - r^n)}{1 - r}$

14.(a) Find equation of a line passing through$(-1, 0)$perpendicular to the line $x + y = 1$.

(b) Given $3x + 4y = 24$, find:

(i) Slope

(ii) X-intercept

(iii) Y-intercept

(c) Sketch the graph of $3x + 4y = 24$.